Heist Architect — Multi-Agent Pathfinding Strategy Game

Project Proposal

Artificial Intelligence

Section: 6J

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PROJECT PROPOSAL

1. Introduction

Multi-Agent Pathfinding (MAPF) is the problem of routing multiple agents through a shared environment without collisions. It is an NP-hard problem with real-world applications in robotics, warehouse automation, and autonomous systems.

This project presents Heist Architect, a two-player asymmetric, turn-based strategy game that transforms MAPF into an interactive experience. One player (the *Mastermind*) plans coordinated paths for multiple thieves inside a guarded building, while the other player (the *Warden*) uses strategic reasoning to detect and intercept them.

The game integrates key AI concepts—pathfinding, probabilistic reasoning, and adversarial decision-making—directly into gameplay, making them visible and interactive rather than hidden in the background.

2. Objectives

The main objective is to design and implement a strategy game that demonstrates practical AI techniques in an engaging and understandable way.

Specific objectives include:

- Implement Multi-Agent Pathfinding for 3–4 agents using a conflict-resolution approach.
- Develop a *pathfinding system (A)** for efficient route planning.
- Design a probability-based tracking system for estimating hidden agent positions.
- Implement adversarial decision-making for AI-controlled guards.
- Create a responsive and interactive graphical user interface.

3. Scope of Work

Core Features

- Grid-based environment representing rooms, corridors, and obstacles
- Multi-agent path planning with collision avoidance

- Hidden movement mechanics for thieves
- Virtual Sensor-based detection system (e.g., cameras, motion sensors)
- Guard movement and interception mechanics

AI Components

- A* for pathfinding
- Conflict handling for multiple agents (CBS)
- Probabilistic tracking (Bayesian Network)
- Minimax-based decision-making for AI guards

Methodology:

Grid & Graph Design: Model the building as a grid-based graph where rooms and corridors are nodes/edges. Each cell has properties: walkable, door, sensor type, camera coverage.

A* Implementation: Implement A* with Manhattan distance heuristic on the building graph, supporting time-indexed blocked-cell constraints for CBS replanning.

CBS Implementation: Implement the two-level CBS algorithm. High level manages the constraint tree with conflict detection and branching; low level replans individual agents via A* with accumulated constraints.

CSP Integration: Inject temporal dependency constraints into CBS, blocking agents from restricted cells until precondition agents complete their tasks. Validate extraction window constraints.

Bayesian Tracking Engine: Maintain a probability grid updated each turn via Bayes' theorem on sensor observations, with a movement transition model for between-turn probability propagation.

Minimax Warden AI: Evaluate guard and camera actions 2–3 turns ahead with alpha-beta pruning, scoring based on detection probability and information gain.

Frontend Development: Build React application for rendering for the game map, and WebSocket for real-time algorithm animation streaming from the Python backend.

Integration & Testing: Connect React frontend to Python FastAPI backend via FastAPI and WebSocket. Test all game modes, edge cases, and algorithm correctness.

5. Tools and Technologies

- **Backend:** FastAPI, Uvicorn, WebSockets
- **Frontend:** React, Phaser
- **Game Logic**: Python

6. Expected Outcome

The project will result in a functional strategy game that:

- Demonstrates **multi-agent pathfinding** in an interactive way
- Shows how **probability can be used to reason under uncertainty**
- Illustrates **strategic decision-making using adversarial search**

The system will serve as both an **educational tool** and a **playable game**, helping users understand complex AI concepts through direct interaction.

8. Conclusion

Heist Architect transforms advanced AI concepts into an engaging game environment. By combining pathfinding, probabilistic reasoning, and strategy, the project bridges the gap between theoretical AI and practical understanding.

It provides a clear demonstration of how multiple AI techniques can work together in a cohesive system while remaining accessible to users.